

Het Patel

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Education

B.S. in Computer Science

August 2023 - May 2027

B.A. in Statistics

- **University at Buffalo, State University of New York**

- **GPA:** 3.96/4.0

- **Coursework:** Operating Systems, Distributed Systems, Machine Learning, Compilers, Computer Vision, Image Processing, Regression

Experience

Full-Stack Software Developer Intern - ATHLYNK

Omaha, NE (Remote)

Tools: React Native, FastAPI, Python, TypeScript, Supabase, PostgreSQL

August 2025 - October 2025

- Engineered end-to-end mobile features using React Native (frontend) and FastAPI (backend), integrating UI components with scalable API services.
- Implemented a real-time messaging system with WebSockets, enabling seamless user-to-user communication.
- Designed and deployed RESTful API endpoints and optimized database CRUD operations with Supabase and PostgreSQL to support dynamic content.
- Improved application reliability and performance by contributing to Agile development cycles, code reviews, and cross-functional integration.

Undergraduate Teaching Assistant - Computer Organization

Buffalo, NY

Tools: MIPS Assembly, VHDL, QtSpim, RISC-V Architecture

January 2026 - Present

- Assist in teaching core computer architecture topics including MIPS assembly, datapath design, pipelining, memory hierarchy, and ALU design.
- Support students in debugging and optimizing MIPS programs using QtSpim and simulating digital circuits in VHDL on EDA Playground.
- Lead office hours and review sessions, and evaluate assignments and provide technical feedback on assembly programming, processor design, and hardware simulation projects.

Technologies

Languages: C, Go, Python, Java, Kotlin, OCaml, JavaScript, TypeScript

Frameworks: React Native, Express.js, FastAPI, Tailwind, Numpy, Pandas, scikit-learn, TensorFlow

Tools: Git, Docker, gdb, Expo, npm, REST APIs, Socket Programming, Minitab

Database: PostgreSQL, Supabase

Environments: Ubuntu (Linux), Windows, Node.js

Projects

Pintos Operating System (Stanford)

[github](#) [🔗](#)

Tools: C, gdb, Makefile

- Implemented threading component of the OS kernel including synchronization primitives and thread scheduling algorithm such as Multilevel Feedback Queue Scheduler (MLFQS).
- Built system call interface including argument validation, safe user memory access, and kernel-user mode transitions, system calls, and process control logic.

CrowdSense

[demo](#) [🔗](#)

Tools: Ultralytics YOLO, Pytorch, OpenCV, FastAPI, Websockets, React Native, Expo

- Built a real-time crowd monitoring system using Ultralytics YOLO model to detect occupancy levels from CCTV camera feeds.
- Designed a multi-service architecture integrating a Python model pipeline, FastAPI backend, and React Native mobile app using WebSockets.